

Mason Scott

240-586-0606 | masonscott141@gmail.com | [linkedin.com/in/mason-t-scott](https://www.linkedin.com/in/mason-t-scott) | github.com/04mScott | masonscott.net

EDUCATION

University of Maryland

Bachelor of Science in Computer Science – Data Science Track, Minor in Statistics

College Park, MD

May 2026

EXPERIENCE

Experio AI

Data & AI Engineering Intern

Bethesda, MD

June 2025 – November 2025

- Designed and built structured data extraction pipelines to parse, normalize, and organize large volumes of unstructured RFP documents into queryable, analytics-ready formats.
- Modeled domain knowledge as a graph database in Neo4j, defining entity schemas and relationships to enable structured querying and reporting across proposal data.
- Developed multi-stage document processing workflows (extraction, validation, transformation, refinement) to produce clean, structured outputs from noisy source documents.
- Built and tested a client-facing analytics demo integrating the Neo4j knowledge graph with a proposal generation pipeline; authored comprehensive test suites in a shared production codebase.

High Sierra Pools

Pool Manager

Frederick, MD

May 2024 – Aug. 2024

- Led and supervised a team of lifeguards, organized daily operations, and coordinated emergency responses with EMS/police.

PROJECTS

EngageCS – Student Engagement Analytics Platform | *Python, Dash, Flask, PostgreSQL, Plotly, Pandas, Docker*

- 6-person capstone commissioned by a UMD CS instructor; deployed to production across ~320 students over 39 sessions (Spring 2026)
- Data engineer and analytics lead; designed the PostgreSQL schema, built a Canvas LMS sync pipeline ingesting courses, rosters, assignments, submissions, and grade snapshots via REST API
- Authored an internal analytics library computing attendance rates, participation scores, at-risk flags, grade distributions, topic engagement correlations, and week-over-week trend deltas from structured data
- Built a registry-driven Plotly visualization library and delivered instructor, TA, and student dashboards surfacing risk badges, engagement heatmaps, and participation timelines; system grew 31% beyond original scope with all acceptance tests signed off

Music Recommendation Engine | *FastAPI, Celery, Redis, MySQL, Docker, Pandas, NumPy*

- Built an async feature pipeline that fetches a user's Spotify library, downloads audio, extracts acoustic embeddings via librosa, and stores feature vectors in MySQL for similarity search
- Implemented cosine similarity over audio feature vectors to generate personalized track recommendations; designed the full MySQL schema for user data, track metadata, and recommendation outputs
- Engineered background task processing with Celery and Redis to handle batch song featurization, YouTube API lookups, and exponential backoff on rate-limited external APIs

Air Quality Predictor | *Python, TensorFlow, Streamlit, AWS, MySQL, Pandas, Plotly*

- Built an automated data pipeline ingesting real-time pollutant readings from the OpenWeatherMap API into a MySQL database on AWS, with scheduled model retraining to incorporate new observations
- Trained an LSTM time-series forecasting model on historical air quality data (CO, NO₂, O₃, PM2.5, PM10); achieved test RMSE of 0.1668; deployed a Streamlit dashboard for multi-city monitoring and trend visualization

TECHNICAL SKILLS

Languages: Python, SQL, R, SAS, Java, C, Rust

Data & Analytics: Pandas, NumPy, Scikit-learn, Plotly, Matplotlib, SQLAlchemy, Tableau

Databases & Storage: PostgreSQL, MySQL, MongoDB, Redis/Valkey, Neo4j

Frameworks & Tools: FastAPI, Dash, Flask, Streamlit, TensorFlow, Docker, AWS, Git, Linux/Unix

Web: React, JavaScript, HTML/CSS